

Who gets flagged?

A CRISP-DM fairness audit of 30-day readmission prediction
UCI Diabetes 130-US Hospitals

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Machine Learning and Data Mining · University of Bologna · A.Y. 2026/27

Individual assignment · Option 2 (CRISP-DM)

The question: who gets the few follow-up slots?

1 Rank

every discharged diabetic inpatient by predicted 30-day readmission risk



2 Flag the top q %

q = training-fold readmission rate (overall **0.114**): a fixed follow-up capacity



3 Audit the errors

equal TPR and FPR across sex, age band and race (equalized odds)

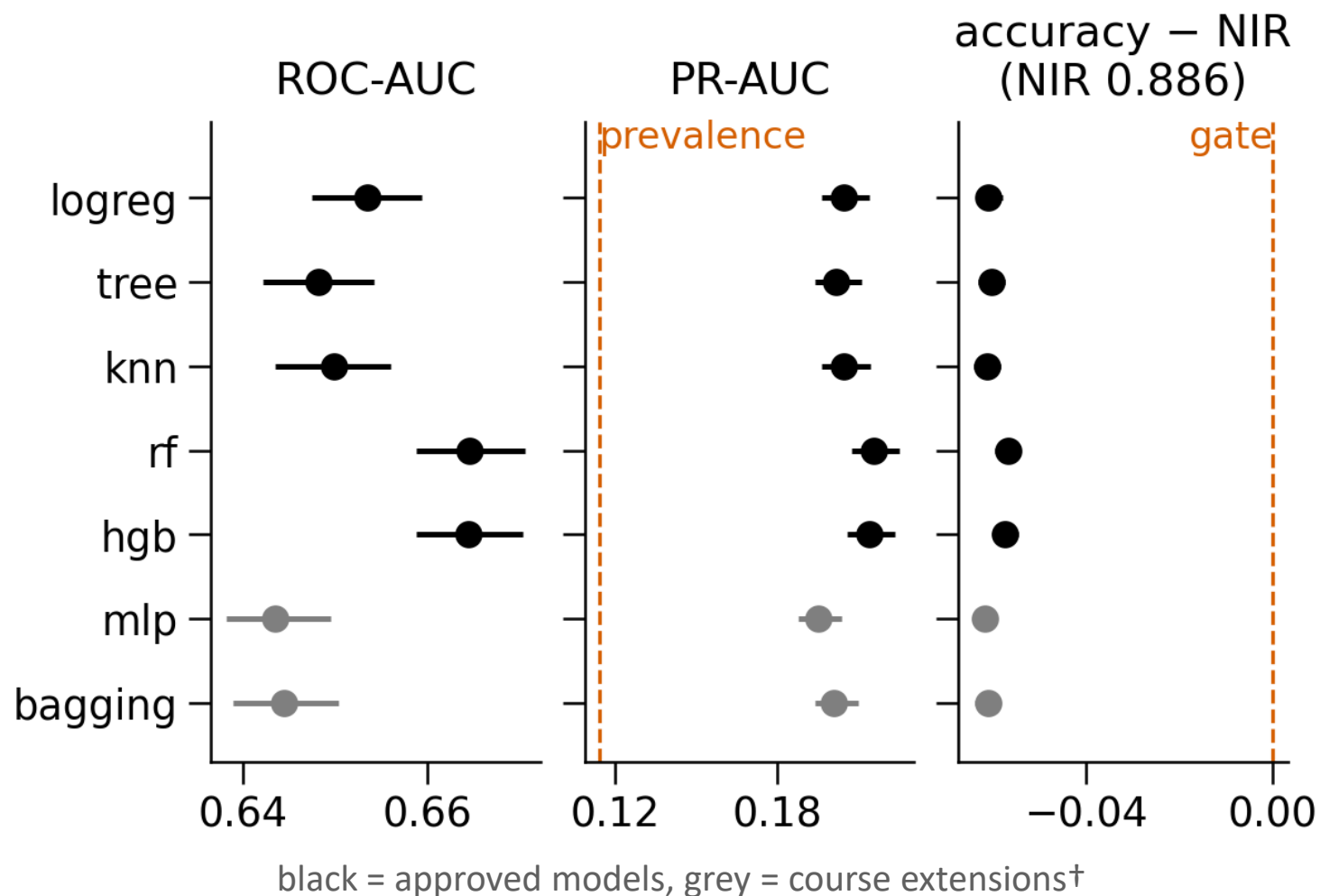
Usability gate (fixed in the approved proposal, before the approved protocol was run): usable only if the lower 95 % bound of accuracy – NIR (0.886) is > 0 .

With m flagged, accuracy – NIR = $(2 \cdot \text{TP} - m) / n$, so the gate needs **PPV > 0.5** : most flags must be right.

Data and pipeline: CRISP-DM, validated patient by patient

Business understanding	flag the top q % at a fixed capacity; equal errors by sex, age band, race
Data understanding	98,470 encounters · 69,303 patients · 11,266 readmitted in 30 days (0.114)
Data preparation	23 inputs in 6 blocks; encoding and scaling fitted on training rows only
Modelling	tree, k-NN, logistic regression, random forest, gradient boosting; depth and k by the one-SE rule
Evaluation	patient-level stratified 5-fold CV × 5 repeats; 1,000 bootstrap draws
Deployment	a recommendation for the hospital, with its limits (slide 9)

Models rank better than chance, but none passes the usability gate

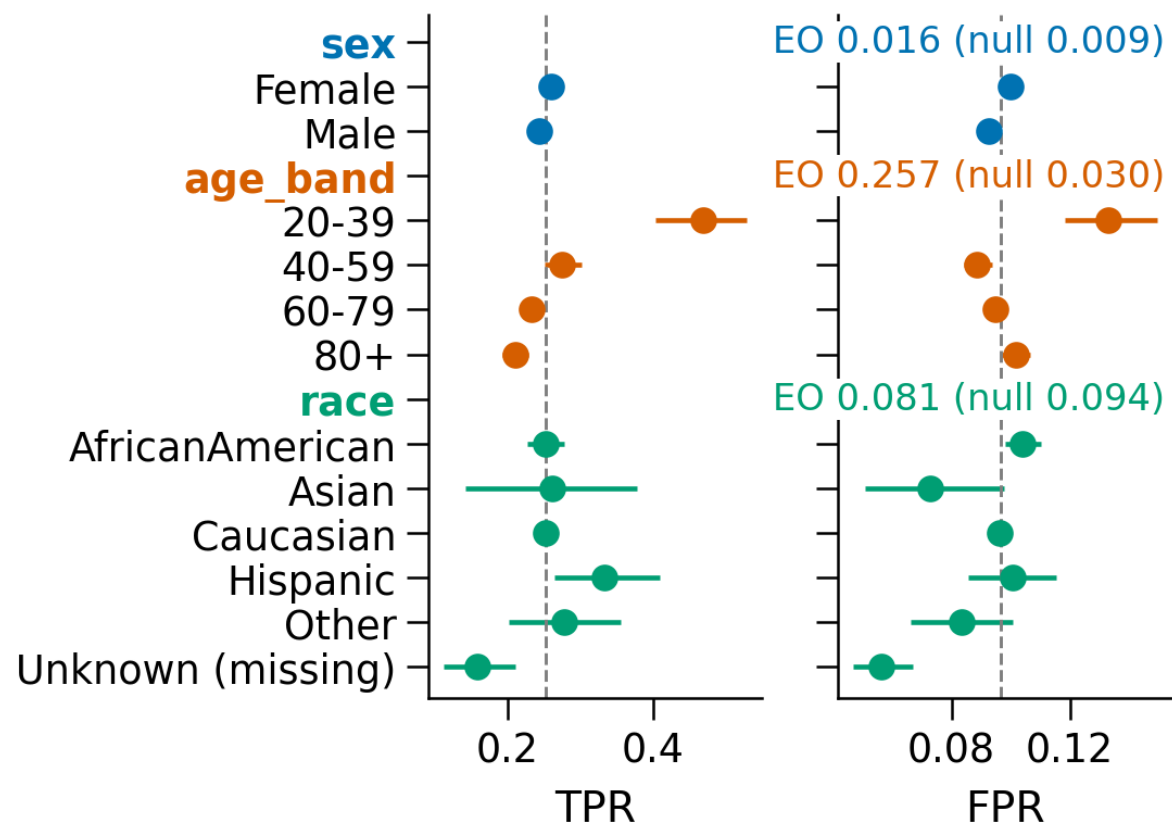


- Random forest: ROC-AUC **0.665** [0.659, 0.671]
- PR-AUC **0.216**, 1.9× the prevalence (0.114)
- At capacity: TPR = PPV = **0.252**
- Accuracy – NIR **-0.057** [-0.059, -0.054]
- Closest rule (top 5 %): PPV 0.307, still below 0.5

**Gate fails: 0 of 34 approved
model × rule pairs pass
(extensions fail too)**

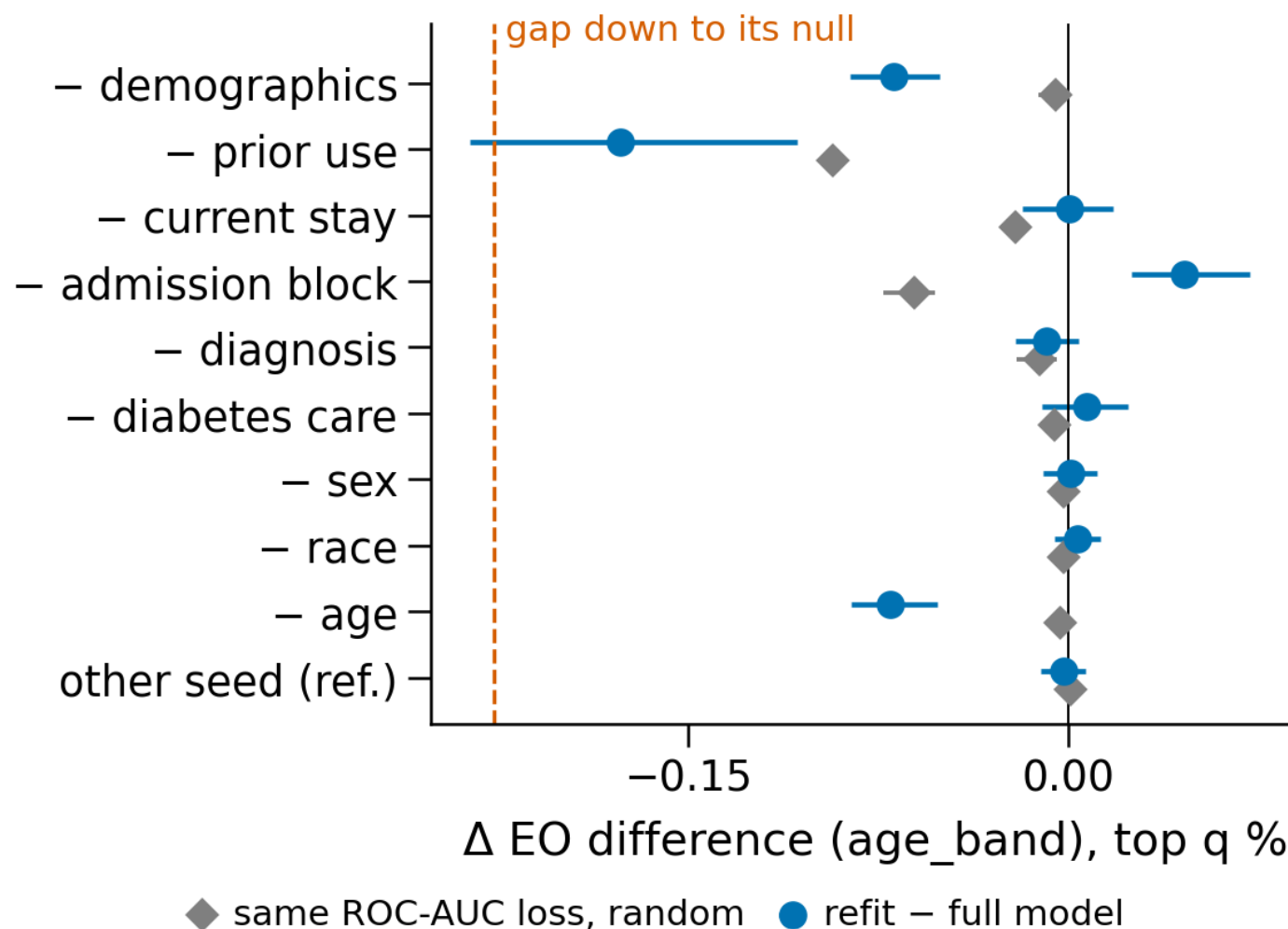
The errors differ by age, not clearly by sex or race

rf, top q %; 95 % CI; hollow = insufficient group; dashed = overall



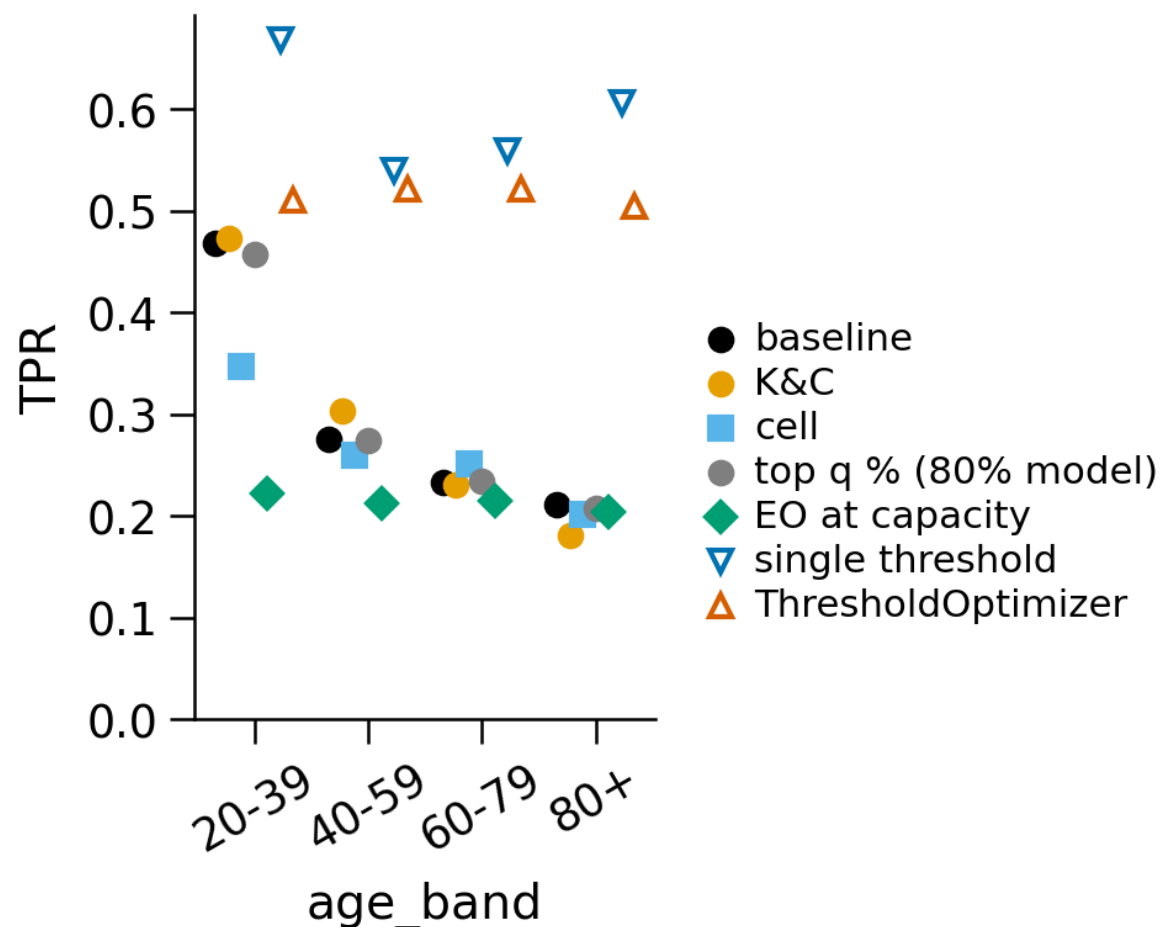
- Age band: EO difference **0.257** [0.192, 0.320]; permutation null† 0.030 (p = 0.005, the minimum for 200 shuffles)
- TPR **0.468** at 20–39 vs **0.211** at 80+; the young are flagged more (0.174 vs 0.107–0.115)
- Sex 0.016 (null 0.009); race 0.081 (null 0.094; “Unknown” = missing code, left out; 0.175 with it): within chance
- Not base rates (0.101–0.125), not calibration (observed – predicted within ± 0.004)

The age gap comes mainly from ranking young patients better



- Ranked within their own group[†], at the overall FPR: TPR **0.383** (20–39) vs **0.201** (80+), most of the 0.257 gap; the shared cut-off adds 0.085 to the young (ROC-AUC 0.755 vs 0.616)
- Without prior-use counts: 0.257 → **0.080** (same AUC loss at random[†]: 0.164); it works by lowering 20–39 TPR (0.468 → 0.245): no remedy
- Without age: 0.187; other inputs predict age band (ROC-AUC 0.773)

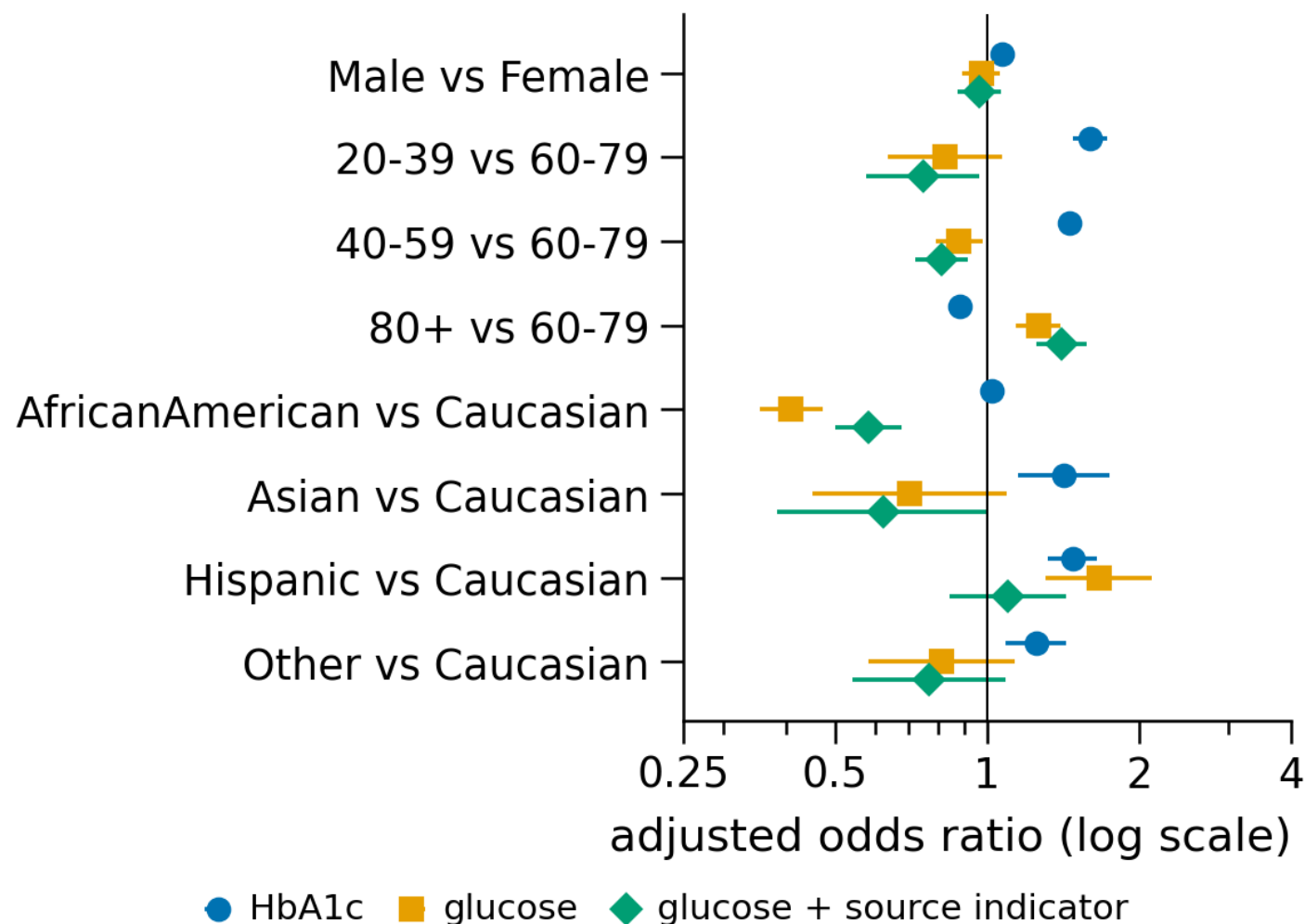
The mitigations that close the gap do so by levelling down



- K&C reweighing widens it: $0.257 \rightarrow 0.293$
- Cell balancing: $0.257 \rightarrow 0.146$ (-0.111); balancing race cells cuts it more (-0.124): mostly class balancing
- EO at capacity[†] (vs same 80 % model, same number flagged): $0.250 \rightarrow 0.018$; TPR 20–39 $0.457 \rightarrow 0.222$, 80+ $0.207 \rightarrow 0.204$
- ThresholdOptimizer (vs one Youden threshold; both flag $\approx 36\%$): $0.188 \rightarrow 0.018$; every age band falls (20–39 $0.668 \rightarrow 0.510$)

Neither post-processor raises any age band's TPR: levelling down, not a fix

Glucose race gaps are largely, not wholly, a recording pattern



- **79.7 %** of glucose results come from the **10.2 %** of encounters with unknown admission type
- With that indicator (post hoc sensitivity fit[†], not tested):
- Hispanic excess vanishes: 1.66 → **1.10** [0.84, 1.43]
- African American deficit shrinks but remains: 0.41 → **0.58** [0.50, 0.68]
- Odds ratios describe who has a recorded result; on their own they are not evidence of unequal care

What a hospital should do with this model

1. Do not use it alone

PPV at capacity **0.252**; the gate needs more than 0.5.

2. Report TPR by age band

beside overall performance: **0.468** at 20–39 vs **0.211** at 80+.

3. Equal TPR is a policy choice

at capacity it lowers young patients' TPR and raises no one's: a cost, not a fix.

4. Improve the data

older patients' prior care; complete admission type and lab recording (not tested here).

Limitations, and what I would do next

Limitations

- Hospital and period are not in the public file: standard errors and bootstrap clustered by patient only
- Intervals are conditional on the fitted models
- “Result recorded” is a proxy for “test ordered”
- Method changed mid-project (new proposal approved); admission-type fit post hoc

Next steps

- Validate on data with hospital and period identifiers
- Improve the ranking for older patients (e.g. richer prior-care data) instead of post-processing
- Agree q with the hospital and report the whole capacity curve ($q = 5\text{--}30\%$ already computed)

Take-home: the model ranks better than chance but is not usable alone; its errors differ by age; equalising them by post-processing is levelling down.

Backup: how to reproduce every number

- Five scripts in order: `run_approved` → `run_fairness` → `run_mitigation` → `run_sources` → `run_ordering`; each writes `outputs/reports/approved_*.md` and `outputs/approved/*.json`
- Checkpoints per outer split carry a fingerprint of settings, prediction code and cohort; a mismatch is refused, never mixed
- Colab notebook: fits one outer split live and redraws every table and figure from the saved JSONs
- pytest collects 86 tests for the approved protocol (`tests/test_readmission_*.py`)
- Paper figures (`scripts/make_figures.py`) and these slides (`slides/build_slides.py`) read the same JSONs: no number typed by hand